

NDA Mathematics — Vectors, Probability & Binomial Distribution

Answer Key

1. (c)

Solution: $|\vec{a}| = \sqrt{3^2 + (-2)^2 + 6^2} = \sqrt{9 + 4 + 36} = \sqrt{49} = 7$. So the answer is 7.

2. (b)

Solution: $|\vec{a}| = \sqrt{4^2 + (-3)^2} = \sqrt{16 + 9} = \sqrt{25} = 5$. The unit vector is $\frac{\vec{a}}{|\vec{a}|} = \frac{1}{5}(4\hat{i} - 3\hat{j})$. So the answer is $\frac{1}{5}(4\hat{i} - 3\hat{j})$.

3. (b)

Solution: $|\vec{a}| = \sqrt{2^2 + 1^2 + 2^2} = \sqrt{9} = 3$. A vector of magnitude 9 along $\vec{a}$ is $9 \cdot \frac{\vec{a}}{3} = 3\vec{a} = 6\hat{i} + 3\hat{j} + 6\hat{k}$. So the answer is $6\hat{i} + 3\hat{j} + 6\hat{k}$.

4. (c)

Solution: $|\vec{a}|^2 |\vec{b}|^2 = (\vec{a} \cdot \vec{b})^2 + |\vec{a} \times \vec{b}|^2$. So $(5)^2 (13)^2 = (\vec{a} \cdot \vec{b})^2 + 25^2$, giving $4225 = (\vec{a} \cdot \vec{b})^2 + 625$, so $(\vec{a} \cdot \vec{b})^2 = 3600$ and $\vec{a} \cdot \vec{b} = 60$. So the answer is 60.

5. (c)

Solution: $|\vec{a} + \vec{b}|^2 + |\vec{a} - \vec{b}|^2 = 2(|\vec{a}|^2 + |\vec{b}|^2) = 2(1 + 1) = 4$. Since $|\vec{a} + \vec{b}|^2 = 3$, we get $|\vec{a} - \vec{b}|^2 = 4 - 3 = 1$, so $|\vec{a} - \vec{b}| = 1$. So the answer is 1.

6. (c)

Solution: $\vec{a} \cdot \vec{b} = (2)(1) + (3)(-1) + (-1)(2) = 2 - 3 - 2 = -3$. So the answer is -3.

7. (a)

Solution: $\cos \theta = \frac{(\hat{i} + \hat{j}) \cdot (\hat{j} + \hat{k})}{|\hat{i} + \hat{j}| |\hat{j} + \hat{k}|} = \frac{0 + 1 + 0}{\sqrt{2} \cdot \sqrt{2}} = \frac{1}{2}$. So $\theta = 60^\circ$. So the answer is 60° .

8. (d)

Solution: $\vec{a} \cdot \vec{b} = (1)(3) + (2)(0) + (2)(4) = 3 + 8 = 11$. $|\vec{a}| = \sqrt{1 + 4 + 4} = 3$, $|\vec{b}| = \sqrt{9 + 0 + 16} = 5$. $\cos \theta = \frac{11}{3 \cdot 5} = \frac{11}{15}$. So the answer is $\frac{11}{15}$.

9. (a)

Solution: For perpendicular vectors $\vec{a} \cdot \vec{b} = 0$: $(2)(1) + (\lambda)(-2) + (1)(3) = 0$, so $2 - 2\lambda + 3 = 0$, giving $5 = 2\lambda$ and $\lambda = \frac{5}{2}$. So the answer is $\frac{5}{2}$.

10. (b)

Solution: $|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2|\vec{a}||\vec{b}| \cos 60^\circ = 9 + 16 + 2(3)(4) \cdot \frac{1}{2} = 25 + 12 = 37$. So $|\vec{a} + \vec{b}| = \sqrt{37}$. So the answer is $\sqrt{37}$.

11. (a)

Solution: $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \frac{3}{2 \cdot 3} = \frac{1}{2}$. So $\theta = 60^\circ$. So the answer is 60° .

12. (c)

Solution: From $\vec{a} + \vec{b} = -\vec{c}$, $|\vec{a} + \vec{b}|^2 = |\vec{c}|^2$: $9 + 25 + 2\vec{a} \cdot \vec{b} = 49$, so $2\vec{a} \cdot \vec{b} = 15$ and $\vec{a} \cdot \vec{b} = \frac{15}{2}$. Then $\cos \theta = \frac{15/2}{3 \cdot 5} = \frac{1}{2}$, so $\theta = 60^\circ$. So the answer is 60° .

13. (b)

Solution: $|\vec{a} - \vec{b}|^2 = 1 + 1 - 2 \cos \theta = 2(1 - \cos \theta) = 4 \sin^2 \frac{\theta}{2}$. So $|\vec{a} - \vec{b}| = 2 \sin \frac{\theta}{2}$ and $\frac{1}{2} |\vec{a} - \vec{b}| = \sin \frac{\theta}{2}$. So the answer is $\sin \frac{\theta}{2}$.

14. (a)

Solution: Scalar projection = $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$. $\vec{a} \cdot \vec{b} = (2)(1) + (3)(2) + (6)(2) = 2 + 6 + 12 = 20$, $|\vec{b}| = \sqrt{1 + 4 + 4} = 3$. Projection = $\frac{20}{3}$. So the answer is $\frac{20}{3}$.

15. (b)

Solution: Vector projection = $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \vec{b}$. $\vec{a} \cdot \vec{b} = 3 + 1 = 4$, $|\vec{b}|^2 = 2$. Projection = $\frac{4}{2}(\hat{i} + \hat{j}) = 2\hat{i} + 2\hat{j}$. So the answer is $2\hat{i} + 2\hat{j}$.

16. (b)

Solution: Projection = $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{(1)(2) + (\lambda)(1) + (1)(2)}{3} = \frac{4 + \lambda}{3} = 2$. So $4 + \lambda = 6$, giving $\lambda = 2$. So the answer is 2.

17. (c)

Solution: $|\vec{a}|^2 = 3^2 + 1^2 + (-1)^2 = 9 + 1 + 1 = 11$. The scalar projection on $\vec{b}$ is $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$, where $\vec{a} \cdot \vec{b} = (3)(1) + (1)(2) + (-1)(2) = 3 + 2 - 2 = 3$ and $|\vec{b}| = \sqrt{1 + 4 + 4} = 3$, so the parallel-component magnitude is $\frac{3}{3} = 1$. The perpendicular component magnitude is $\sqrt{|\vec{a}|^2 - (\text{parallel})^2} = \sqrt{11 - 1} = \sqrt{10}$. So the answer is $\sqrt{10}$.

18. (d)

Solution: $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = \hat{i}(1 \cdot 1 - 0 \cdot 1) - \hat{j}(1 \cdot 1 - 0 \cdot 0) + \hat{k}(1 \cdot 1 - 1 \cdot 0) = \hat{i} - \hat{j} + \hat{k}$. So the answer is $\hat{i} - \hat{j} + \hat{k}$.

19. (d)

Solution: Area = $|\vec{a} \times \vec{b}| = |3\hat{i} \times 2\hat{j}| = |6(\hat{i} \times \hat{j})| = |6\hat{k}| = 6$. So the answer is 6.

20. (d)

Solution: $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin 30^\circ = 4 \cdot 5 \cdot \frac{1}{2} = 10$.
So the answer is 10.

21. (a)

Solution: $\vec{AB} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{AC} = 3\hat{i} - \hat{j} + \hat{k}$.
 $\vec{AB} \times \vec{AC} = \hat{i}(2 \cdot 1 - 3 \cdot (-1)) - \hat{j}(1 \cdot 1 - 3 \cdot 3) + \hat{k}(1 \cdot (-1) - 2 \cdot 3) = \hat{i}(2 + 3) - \hat{j}(1 - 9) + \hat{k}(-1 - 6) = 5\hat{i} + 8\hat{j} - 7\hat{k}$. Magnitude = $\sqrt{25 + 64 + 49} = \sqrt{138}$. Area of triangle = $\frac{1}{2} |\vec{AB} \times \vec{AC}| = \frac{\sqrt{138}}{2}$. So the answer is $\frac{\sqrt{138}}{2}$.

22. (b)

Solution: $\vec{a} \times \vec{b} = \hat{i} - \hat{j} + \hat{k}$ (from the standard determinant), with magnitude $\sqrt{1 + 1 + 1} = \sqrt{3}$.
The unit perpendicular vector is $\frac{1}{\sqrt{3}}(\hat{i} - \hat{j} + \hat{k})$. So the answer is $\frac{1}{\sqrt{3}}(\hat{i} - \hat{j} + \hat{k})$.

23. (c)

Solution: Area of a parallelogram with diagonals $\vec{d}_1, \vec{d}_2$ is $\frac{1}{2} |\vec{d}_1 \times \vec{d}_2|$. $\vec{a} \times \vec{b} = \hat{i}(1 \cdot 1 - (-3)(-2)) - \hat{j}(2 \cdot 1 - (-3)(1)) + \hat{k}(2 \cdot (-2) - 1 \cdot 1) = \hat{i}(1 - 6) - \hat{j}(2 + 3) + \hat{k}(-4 - 1) = -5\hat{i} - 5\hat{j} - 5\hat{k}$.
Magnitude = $5\sqrt{3}$. Area = $\frac{1}{2}(5\sqrt{3}) = \frac{5\sqrt{3}}{2}$. So the answer is $\frac{5\sqrt{3}}{2}$.

24. (b)

Solution: $\vec{a} \times \vec{b} = \vec{c} \times \vec{b}$ gives $(\vec{a} - \vec{c}) \times \vec{b} = \vec{0}$.
This means $\vec{a} - \vec{c}$ is parallel to $\vec{b}$, i.e. $\vec{a} - \vec{c} = t\vec{b}$ for some scalar t . So the answer is $\vec{a} - \vec{c} = t\vec{b}$ for some scalar t .

25. (b)

Solution: $[\hat{i}\hat{j}\hat{k}] = \hat{i} \cdot (\hat{j} \times \hat{k}) = \hat{i} \cdot \hat{i} = 1$. So the answer is 1.

26. (d)

Solution: $[\vec{a}\vec{b}\vec{c}] = \begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix} = 1(1 - 0) - 1(0 - 1) + 0 = 1 + 1 = 2$. The volume is $|2| = 2$. So the answer is 2.

27. (b)

Solution: $[\vec{a}\vec{b}\vec{c}] = \begin{vmatrix} 2 & -3 & 0 \\ 1 & 1 & -1 \\ 3 & 0 & -1 \end{vmatrix} = 2(1 \cdot (-1) - (-1) \cdot 0) - (-3)(1 \cdot (-1) - (-1) \cdot 3) + 0 = 2(-1) + 3(-1 + 3) = -2 + 3(2) = -2 + 6 = 4$.
So the answer is 4.

28. (b)

Solution: Coplanarity requires $[\vec{a}\vec{b}\vec{c}] = 0$:
 $\begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & \lambda & -1 \end{vmatrix} = 1(3 \cdot (-1) - 1 \cdot \lambda) - 1(2 \cdot (-1) - 1 \cdot 1) + 1(2 \cdot \lambda - 3 \cdot 1)$. This equals $(-3 - \lambda) -$

$(-2 - 1) + (2\lambda - 3) = -3 - \lambda + 3 + 2\lambda - 3 = \lambda - 3$. Setting $\lambda - 3 = 0$ gives $\lambda = 3$. So the answer is 3.

29. (a)

Solution: A standard identity gives $[\vec{a} + \vec{b}\vec{b} + \vec{c}\vec{c} + \vec{a}] = 2[\vec{a}\vec{b}\vec{c}]$. So the value is $2 \cdot 5 = 10$. So the answer is 10.

30. (c)

Solution: $\vec{AB} = \hat{i} + 2\hat{j} + 2\hat{k}$, $\vec{AC} = -\hat{i} + \hat{j} + 3\hat{k}$,
 $\vec{AD} = 2\hat{i} + 3\hat{j} + 6\hat{k}$. $[\vec{AB}\vec{AC}\vec{AD}] = \begin{vmatrix} 1 & 2 & 2 \\ -1 & 1 & 3 \\ 2 & 3 & 6 \end{vmatrix} = 1(1 \cdot 6 - 3 \cdot 3) - 2((-1) \cdot 6 - 3 \cdot 2) + 2((-1) \cdot 3 - 1 \cdot 2) = 1(6 - 9) - 2(-6 - 6) + 2(-3 - 2) = -3 + 24 - 10 = 11$. Volume = $|11| = 11$. So the answer is 11.

31. (a)

Solution: Parallel vectors have proportional components: $\frac{2}{3} = \frac{4}{6} = \frac{\lambda}{9}$. From $\frac{2}{3} = \frac{\lambda}{9}$, $\lambda = \frac{2 \cdot 9}{3} = 6$. So the answer is 6.

32. (a)

Solution: Perpendicular: $\vec{a} \cdot \vec{b} = 0$: $4\mu + 2(-9) + 1 \cdot 2 = 0$, so $4\mu - 18 + 2 = 0$, giving $4\mu = 16$ and $\mu = 4$. So the answer is 4.

33. (d)

Solution: $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = |\vec{a}|^2 - |\vec{b}|^2 = 0$. So $|\vec{a}|^2 = |\vec{b}|^2$, giving $|\vec{a}| = |\vec{b}|$. So the answer is $|\vec{a}| = |\vec{b}|$.

34. (a)

Solution: Require $(\vec{a} + \lambda\vec{b}) \cdot \vec{a} = 0$: $|\vec{a}|^2 + \lambda(\vec{b} \cdot \vec{a}) = 0$. $|\vec{a}|^2 = 1 + 4 + 9 = 14$, $\vec{b} \cdot \vec{a} = 2 - 2 + 3 = 3$. So $14 + 3\lambda = 0$, giving $\lambda = -\frac{14}{3}$. So the answer is $-\frac{14}{3}$.

35. (b)

Solution: Perpendicular: $(2)(1) + (-1)(p) + (1)(-3) = 0$, so $2 - p - 3 = 0$, giving $-p - 1 = 0$ and $p = -1$. So the answer is -1.

36. (c)

Solution: Magnitude = $\sqrt{4 + 1 + 4} = 3$. Direction cosines are $\frac{2}{3}, -\frac{1}{3}, \frac{2}{3}$. So the answer is $\frac{2}{3}, -\frac{1}{3}, \frac{2}{3}$.

37. (a)

Solution: Direction cosines satisfy $l^2 + m^2 + n^2 = 1$: $\frac{1}{4} + \frac{1}{4} + n^2 = 1$, so $n^2 = \frac{1}{2}$ and $n = \frac{1}{\sqrt{2}}$. So the answer is $\frac{1}{\sqrt{2}}$.

38. (d)

Solution: With equal angles, $l = m = n = \cos \alpha$. Then $3 \cos^2 \alpha = 1$, so $\cos \alpha = \frac{1}{\sqrt{3}}$ (taking the positive acute value). So the answer is $\frac{1}{\sqrt{3}}$.

39. (d)

Solution: $\cos \theta = \frac{(1)(2)+(-2)(2)+(2)(1)}{\sqrt{1+4+4}\sqrt{4+4+1}} = \frac{2-4+2}{3 \cdot 3} = \frac{0}{9} = 0$. So $\theta = 90^\circ$. So the answer is 90° .

40. (d)

Solution: Midpoint position vector $= \frac{1}{2}((2\hat{i} + 3\hat{j}) + (4\hat{i} - \hat{j})) = \frac{1}{2}(6\hat{i} + 2\hat{j}) = 3\hat{i} + \hat{j}$. So the answer is $3\hat{i} + \hat{j}$.

41. (b)

Solution: Section formula (ratio $m:n = 2:1$): $\frac{m\vec{B}+n\vec{A}}{m+n} = \frac{2\vec{B}+1\vec{A}}{3}$. $2\vec{B} = 8\hat{i} + 10\hat{j} + 12\hat{k}$; adding $\vec{A}$ gives $9\hat{i} + 12\hat{j} + 15\hat{k}$; dividing by 3 gives $3\hat{i} + 4\hat{j} + 5\hat{k}$. So the answer is $3\hat{i} + 4\hat{j} + 5\hat{k}$.

42. (c)

Solution: For collinearity, $\vec{AB}$ and $\vec{AC}$ must be parallel. $\vec{AB} = \hat{i} + 2\hat{j}$, $\vec{AC} = 2\hat{i} + (\lambda - 1)\hat{j}$. Proportional: $\frac{1}{2} = \frac{2}{\lambda - 1}$, so $\lambda - 1 = 4$ and $\lambda = 5$. So the answer is 5.

43. (a)

Solution: Centroid $= \frac{1}{3}$ of the sum: $(3 + 1 + 2)\hat{i} + (-1 + 2 + 2)\hat{j} + (2 - 1 + 5)\hat{k} = 6\hat{i} + 3\hat{j} + 6\hat{k}$; dividing by 3 gives $2\hat{i} + \hat{j} + 2\hat{k}$. So the answer is $2\hat{i} + \hat{j} + 2\hat{k}$.

44. (d)

Solution: Let C divide AB in ratio $k:1$. Using the $\hat{i}$ -component of the section formula $\frac{k(8)+1(2)}{k+1} = 4$: $8k + 2 = 4(k + 1) = 4k + 4$, so $4k = 2$ and $k = \frac{1}{2}$. Hence $AC:CB = \frac{1}{2}:1 = 1:2$. So the answer is 1:2.

45. (d)

Solution: Resultant $\vec{a} + \vec{b} = 3\hat{i} + \hat{j} + \hat{k}$, magnitude $= \sqrt{9 + 1 + 1} = \sqrt{11}$. Unit vector $= \frac{1}{\sqrt{11}}(3\hat{i} + \hat{j} + \hat{k})$. So the answer is $\frac{1}{\sqrt{11}}(3\hat{i} + \hat{j} + \hat{k})$.

46. (b)

Solution: The internal bisector is along $\hat{a} + \hat{b}$. $\hat{a} = \frac{3\hat{i}+4\hat{j}}{5}$, $\hat{b} = \frac{5\hat{j}}{5} = \hat{j}$. Sum $= \frac{3}{5}\hat{i} + \frac{4}{5}\hat{j} + \hat{j} = \frac{3}{5}\hat{i} + \frac{9}{5}\hat{j}$, which is proportional to $3\hat{i} + 9\hat{j}$. So the answer is $3\hat{i} + 9\hat{j}$.

47. (d)

Solution: $x(\hat{i} + \hat{j}) + y(\hat{i} - \hat{j}) = (x + y)\hat{i} + (x - y)\hat{j} = 3\hat{i} + 5\hat{j}$. So $x + y = 3$ and $x - y = 5$. Adding: $2x = 8$, $x = 4$; then $y = -1$. So the answer is $(4, -1)$.

48. (c)

Solution: Two non-zero, non-collinear (linearly independent) vectors can combine to the zero vector only with both scalars zero. Hence $x = 0$ and $y = 0$. So the answer is $x = 0$ and $y = 0$.

49. (c)

Solution: Displacement $\vec{d} = (2 - 1)\hat{i} + (2 - 0)\hat{j} + (3 - 1)\hat{k} = \hat{i} + 2\hat{j} + 2\hat{k}$. Work $= \vec{F} \cdot \vec{d} = (2)(1) + (3)(2) + (1)(2) = 2 + 6 + 2 = 10$. So the answer is 10.

50. (c)

Solution: Torque $= \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 2 & -1 & 3 \end{vmatrix} = \hat{i}(1 \cdot 3 - 1 \cdot (-1)) - \hat{j}(1 \cdot 3 - 1 \cdot 2) + \hat{k}(1 \cdot (-1) - 1 \cdot 2) = \hat{i}(3 + 1) - \hat{j}(3 - 2) + \hat{k}(-1 - 2) = 4\hat{i} - \hat{j} - 3\hat{k}$. So the answer is $4\hat{i} - \hat{j} - 3\hat{k}$.

51. (a)

Solution: Favourable outcomes for a sum of 9: $(3,6), (4,5), (5,4), (6,3)$ — that is 4 outcomes out of 36. Probability $= \frac{4}{36} = \frac{1}{9}$.

52. (b)

Solution: Numbers greater than 4 are 5 and 6: 2 outcomes out of 6. Probability $= \frac{2}{6} = \frac{1}{3}$.

53. (a)

Solution: Total outcomes $= 2^3 = 8$. Exactly two heads: $\binom{3}{2} = 3$ ways (HHT, HTH, THH). Probability $= \frac{3}{8}$.

54. (d)

Solution: Total outcomes $= 2^4 = 16$. At least three heads means exactly 3 or exactly 4: $\binom{4}{3} + \binom{4}{4} = 4 + 1 = 5$. Probability $= \frac{5}{16}$.

55. (c)

Solution: There are 4 kings in 52 cards. Probability $= \frac{4}{52} = \frac{1}{13}$.

56. (d)

Solution: Hearts = 13, face cards = 12, hearts that are face cards = 3. By addition theorem: $13 + 12 - 3 = 22$ favourable. Probability $= \frac{22}{52} = \frac{11}{26}$.

57. (c)

Solution: Favourable $= \binom{4}{2} = 6$, total $= \binom{52}{2} = 1326$. Probability $= \frac{6}{1326} = \frac{1}{221}$.

58. (b)

Solution: The product is odd only when both numbers are odd: $3 \times 3 = 9$ outcomes. So even-

product outcomes = $36 - 9 = 27$. Probability = $\frac{27}{36} = \frac{3}{4}$.

59. (c)

Solution: Total outcomes = $6^3 = 216$. All different: $6 \times 5 \times 4 = 120$. Probability = $\frac{120}{216} = \frac{5}{9}$.

60. (b)

Solution: The sum is even in two cases: both even ($3 \times 3 = 9$ outcomes) or both odd ($3 \times 3 = 9$ outcomes), total 18. Favourable (both even) = 9. Probability = $\frac{9}{18} = \frac{1}{2}$.

61. (c)

Solution: Total balls = 8, blue = 3. Probability = $\frac{3}{8}$.

62. (b)

Solution: Favourable = $\binom{6}{2} = 15$, total = $\binom{10}{2} = 45$. Probability = $\frac{15}{45} = \frac{1}{3}$.

63. (a)

Solution: Favourable = $\binom{5}{2}\binom{4}{1} = 10 \times 4 = 40$. Total = $\binom{9}{3} = 84$. Probability = $\frac{40}{84} = \frac{10}{21}$.

64. (a)

Solution: Favourable = $3 \times 4 \times 5 = 60$. Total = $\binom{12}{3} = 220$. Probability = $\frac{60}{220} = \frac{3}{11}$.

65. (d)

Solution: INDIA has 5 letters with the letter I repeated twice, so total distinct arrangements = $\frac{5!}{2!} = 60$. Treating the two I's as one block: $4! = 24$ arrangements. Probability = $\frac{24}{60} = \frac{2}{5}$.

66. (d)

Solution: Total ways to select = $\binom{7}{3}\binom{4}{2} = 35 \times 6 = 210$. Only one specific set is favourable, so probability = $\frac{1}{210}$.

67. (d)

Solution: Total arrangements = $5! = 120$. Treating A and B as a block: $4! \times 2! = 24 \times 2 = 48$. Probability = $\frac{48}{120} = \frac{2}{5}$.

68. (d)

Solution: By the addition theorem, $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.5 + 0.3 - 0.1 = 0.7 = \frac{7}{10}$.

69. (a)

Solution: For mutually exclusive events $P(A \cap B) = 0$, so $P(A \cup B) = P(A) + P(B) = \frac{1}{4} + \frac{1}{3} = \frac{3+4}{12} = \frac{7}{12}$.

70. (a)

Solution: Divisible by 3: 10 numbers; by 5: 6 numbers; by 15 (both): 2 numbers. Favourable = $10 + 6 - 2 = 14$. Probability = $\frac{14}{30} = \frac{7}{15}$.

71. (c)

Solution: From $P(A \cup B) = P(A) + P(B) - P(A \cap B)$: $\frac{3}{5} = \frac{2}{5} + \frac{1}{3} - P(A \cap B)$. So $P(A \cap B) = \frac{6}{15} + \frac{5}{15} - \frac{9}{15} = \frac{2}{15}$.

72. (a)

Solution: Spades = 13, kings = 4 (one of which is a spade, already counted, so 3 new), red queens = 2 (disjoint from the above). Favourable = $13 + 3 + 2 = 18$. Probability = $\frac{18}{52} = \frac{9}{26}$.

73. (c)

Solution: Probability of complement = $1 - \frac{2}{7} = \frac{5}{7}$.

74. (c)

Solution: $P(\text{at least one tail}) = 1 - P(\text{no tail}) = 1 - \frac{1}{8} = \frac{7}{8}$, since all heads has probability $\frac{1}{8}$.

75. (a)

Solution: $P(\text{no green}) = \frac{\binom{6}{2}}{\binom{10}{2}} = \frac{15}{45} = \frac{1}{3}$. So $P(\text{at least one green}) = 1 - \frac{1}{3} = \frac{2}{3}$.

76. (a)

Solution: $P(\text{no six in one throw}) = \frac{5}{6}$. $P(\text{no six in three throws}) = \left(\frac{5}{6}\right)^3 = \frac{125}{216}$. So $P(\text{at least one six}) = 1 - \frac{125}{216} = \frac{91}{216}$.

77. (c)

Solution: $P(\text{none solves}) = \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{4}$. So $P(\text{at least one}) = 1 - \frac{1}{4} = \frac{3}{4}$.

78. (a)

Solution: $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1/4}{1/2} = \frac{1}{2}$.

79. (d)

Solution: Even numbers: $\{2, 4, 6\}$, so $P(B) = \frac{3}{6}$. Even and greater than 3: $\{4, 6\}$, so $P(A \cap B) = \frac{2}{6}$. Conditional probability = $\frac{2/6}{3/6} = \frac{2}{3}$.

80. (d)

Solution: After drawing one king, 3 kings remain in 51 cards. Required probability = $\frac{3}{51} = \frac{1}{17}$.

81. (c)

Solution: Sum = 7: $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$ — 6 outcomes.

Those containing a 2: (2,5), (5,2) — 2 outcomes.
Probability = $\frac{2}{6} = \frac{1}{3}$.

82. (b)

Solution: Sample space (with order):
{BB, BG, GB, GG}. Given at least one boy, the
favourable space is {BB, BG, GB}, of which only BB
has both boys. Probability = $\frac{1}{3}$.

83. (d)

Solution: First $P(A \cap B) = P(A) + P(B) - P(A \cup B)$
 $= \frac{3}{8} + \frac{5}{8} - \frac{6}{8} = \frac{2}{8} = \frac{1}{4}$. Then $P(A|B) = \frac{1/4}{5/8} = \frac{1}{4}$.
 $\frac{8}{5} = \frac{2}{5}$.

84. (b)

Solution: For independent events $P(A \cap B) =$
 $P(A) \cdot P(B) = \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{12}$.

85. (c)

Solution: $P(\text{both work}) = 0.8 \times 0.6 = 0.48 =$
 $\frac{48}{100} = \frac{12}{25}$.

86. (a)

Solution: $P(\text{head}) = \frac{1}{2}$ and $P(\text{number} > 4) = \frac{2}{6} = \frac{1}{3}$.
These are independent: $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$.

87. (b)

Solution: For independent events $P(A \cap B) = \frac{2}{5} \cdot$
 $\frac{1}{2} = \frac{1}{5}$. Then $P(A \cup B) = \frac{2}{5} + \frac{1}{2} - \frac{1}{5} = \frac{4}{10} + \frac{5}{10} - \frac{2}{10} =$
 $\frac{7}{10}$.

88. (c)

Solution: Let the events be solving by each. $P(\text{only}$
 $\text{first}) = \frac{1}{3} \cdot \frac{3}{4} \cdot \frac{4}{5} = \frac{12}{60}$; $P(\text{only second}) = \frac{2}{3} \cdot \frac{1}{4} \cdot \frac{4}{5} = \frac{8}{60}$;
 $P(\text{only third}) = \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{1}{5} = \frac{6}{60}$. Sum = $\frac{26}{60} = \frac{13}{30}$.

89. (a)

Solution: For independent events $P(A \cup B) =$
 $P(A) + P(B) - P(A)P(B)$. So $0.65 = 0.3 +$
 $P(B)(1 - 0.3) = 0.3 + 0.7 P(B)$. Then $0.7 P(B) =$
 0.35 , giving $P(B) = 0.5 = \frac{1}{2}$.

90. (a)

Solution: $P(\text{red}) = \frac{1}{2} \cdot \frac{3}{5} + \frac{1}{2} \cdot \frac{2}{6} = \frac{3}{10} + \frac{1}{6} = \frac{9}{30} + \frac{5}{30} =$
 $\frac{14}{30} = \frac{7}{15}$.

91. (b)

Solution: $P(\text{Bag I} \cap \text{red}) = \frac{1}{2} \cdot \frac{3}{5} = \frac{3}{10}$; $P(\text{Bag II} \cap$
 $\text{red}) = \frac{1}{2} \cdot \frac{2}{6} = \frac{1}{6}$. $P(\text{red}) = \frac{9}{30} + \frac{5}{30} = \frac{14}{30}$. By Bayes:
 $\frac{9/30}{14/30} = \frac{9}{14}$.

92. (d)

Solution: $P(\text{defective}) = 0.6 \times 0.02 + 0.4 \times 0.05 =$
 $0.012 + 0.020 = 0.032 = \frac{32}{1000} = \frac{4}{125}$.

93. (b)

Solution: $P(B \cap \text{def}) = 0.4 \times 0.05 = 0.020$; $P(A \cap$
 $\text{def}) = 0.6 \times 0.02 = 0.012$; $P(\text{def}) = 0.032$. By
Bayes: $\frac{0.020}{0.032} = \frac{20}{32} = \frac{5}{8}$.

94. (b)

Solution: $P(\text{box1} \cap \text{gold}) = \frac{1}{2} \cdot \frac{2}{5} = \frac{2}{10}$; $P(\text{box2} \cap$
 $\text{gold}) = \frac{1}{2} \cdot \frac{4}{5} = \frac{4}{10}$. $P(\text{gold}) = \frac{6}{10}$. By Bayes: $\frac{4/10}{6/10} =$
 $\frac{4}{6} = \frac{2}{3}$.

95. (d)

Solution: $P(\text{pos} \cap \text{disease}) = 0.01 \times 0.99 =$
 0.0099 ; $P(\text{pos} \cap \text{healthy}) = 0.99 \times 0.02 = 0.0198$;
 $P(\text{pos}) = 0.0099 + 0.0198 = 0.0297$. By Bayes:
 $\frac{0.0099}{0.0297} = \frac{99}{297} = \frac{1}{3}$.

96. (d)

Solution: If odds in favour are 3: 5, probability =
 $\frac{3}{3+5} = \frac{3}{8}$.

97. (b)

Solution: $P(\text{event}) = \frac{2}{7}$ so $P(\text{not event}) = \frac{5}{7}$. Odds
against = $\frac{P(\text{not})}{P(\text{event})} = 5:2$.

98. (d)

Solution: A leap year has $366 = 52 \times 7 + 2$ days,
so 52 Sundays plus 2 extra days. The two extra
days are one of 7 equally likely consecutive pairs;
2 of these pairs contain a Sunday (Sat-Sun, Sun-
Mon). Probability = $\frac{2}{7}$.

99. (b)

Solution: A non-leap year has $365 = 52 \times 7 + 1$
days, so 52 Mondays plus 1 extra day. The extra
day is equally likely to be any of the 7 days; only 1
of them is Monday. Probability = $\frac{1}{7}$.

100. (c)

Solution: Total ways = $7^3 = 343$. All different
days: $7 \times 6 \times 5 = 210$. Probability = $\frac{210}{343} = \frac{30}{49}$.

101. (a)

Solution: Here $n = 5$, $p = \frac{1}{2}$, $q = \frac{1}{2}$. Then $P(X =$
 $3) = \binom{5}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 = 10 \cdot \frac{1}{32} = \frac{10}{32} = \frac{5}{16}$.

102. (b)

Solution: Here $n = 4$, $p = \frac{1}{6}$, $q = \frac{5}{6}$. Then $P(X =$
 $2) = \binom{4}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^2 = 6 \cdot \frac{1}{36} \cdot \frac{25}{36} = \frac{150}{1296} = \frac{25}{216}$.

103. (b)

Solution: Probability defective $p = 0.2 = \frac{1}{5}$, so $q = \frac{4}{5}$, with $n = 5$. Then $P(X = 2) = \binom{5}{2} \left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^3$, which is the required form.

104. (d)

Solution: Here $n = 4, p = \frac{3}{4}, q = \frac{1}{4}$. Then $P(X = 3) = \binom{4}{3} \left(\frac{3}{4}\right)^3 \left(\frac{1}{4}\right)^1 = 4 \cdot \frac{27}{64} \cdot \frac{1}{4} = \frac{108}{256} = \frac{27}{64}$.

105. (a)

Solution: Here $n = 6, p = \frac{3}{5}, q = \frac{2}{5}$. Then $P(X = 4) = \binom{6}{4} \left(\frac{3}{5}\right)^4 \left(\frac{2}{5}\right)^2 = 15 \cdot \frac{81}{625} \cdot \frac{4}{25} = \frac{15 \times 324}{15625} = \frac{4860}{15625}$.

106. (c)

Solution: Here $n = 18, p = \frac{1}{6}$. Mean $= np = 18 \times \frac{1}{6} = 3$.

107. (d)

Solution: Variance $= npq = 12 \times \frac{1}{4} \times \frac{3}{4} = \frac{36}{16} = \frac{9}{4}$.

108. (d)

Solution: Mean $np = 4$ and variance $npq = \frac{4}{3}$. Dividing, $q = \frac{npq}{np} = \frac{4/3}{4} = \frac{1}{3}$, so $p = \frac{2}{3}$. Then $n = \frac{4}{p} = \frac{4}{2/3} = 6$.

109. (a)

Solution: Mean $np = 6$, variance $npq = 4$. Then $q = \frac{npq}{np} = \frac{4}{6} = \frac{2}{3}$, so $p = 1 - q = \frac{1}{3}$.

110. (a)

Solution: Mean $np = 20$ and variance $npq = \sigma^2 = 4^2 = 16$. Dividing, $q = \frac{npq}{np} = \frac{16}{20} = \frac{4}{5}$, so $p = \frac{1}{5}$. Then $n = \frac{20}{p} = \frac{20}{1/5} = 100$.

111. (c)

Solution: Here $n = 4, p = \frac{1}{6}, q = \frac{5}{6}$. $P(\text{at least one six}) = 1 - P(X = 0) = 1 - \left(\frac{5}{6}\right)^4 = 1 - \frac{625}{1296} = \frac{671}{1296}$.

112. (b)

Solution: Here $n = 3, p = \frac{1}{2}$. $P(\text{at least one head}) = 1 - P(X = 0) = 1 - \left(\frac{1}{2}\right)^3 = 1 - \frac{1}{8} = \frac{7}{8}$.

113. (c)

Solution: Here $n = 3, p = \frac{1}{10}, q = \frac{9}{10}$. $P(X \leq 1) = P(X = 0) + P(X = 1) = \left(\frac{9}{10}\right)^3 + \binom{3}{1} \left(\frac{1}{10}\right) \left(\frac{9}{10}\right)^2 = \frac{729}{1000} + 3 \cdot \frac{81}{1000} = \frac{729+243}{1000} = \frac{972}{1000}$.

114. (d)

Solution: Here $n = 5, p = \frac{1}{3}, q = \frac{2}{3}$. $P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 1 - \left(\frac{2}{3}\right)^5 - \binom{5}{1} \left(\frac{1}{3}\right) \left(\frac{2}{3}\right)^4 = 1 - \frac{32}{243} - 5 \cdot \frac{16}{243} = 1 - \frac{32}{243} - \frac{80}{243} = \frac{243-112}{243} = \frac{131}{243}$.

115. (b)

Solution: Here $n = 7, p = \frac{1}{2}$. The mode lies in $(n + 1)p - 1 \leq m \leq (n + 1)p$, i.e. $8 \cdot \frac{1}{2} - 1 = 3 \leq m \leq 4$. Since $(n + 1)p = 4$ is an integer, both 3 and 4 are most probable.

116. (c)

Solution: Here $n = 10, p = \frac{1}{6}$. The mode satisfies $(n + 1)p - 1 \leq m \leq (n + 1)p$, i.e. $11 \cdot \frac{1}{6} - 1 \leq m \leq 11 \cdot \frac{1}{6}$, that is $0.833 \leq m \leq 1.833$. The only integer in this range is $m = 1$.

117. (c)

Solution: Here $n = 9, p = \frac{1}{3}$. The mode satisfies $(n + 1)p - 1 \leq m \leq (n + 1)p$, i.e. $10 \cdot \frac{1}{3} - 1 \leq m \leq 10 \cdot \frac{1}{3}$, that is $2.33 \leq m \leq 3.33$. The only integer is $m = 3$.

118. (a)

Solution: We need $1 - \left(\frac{1}{2}\right)^n > 0.9$, i.e. $\left(\frac{1}{2}\right)^n < 0.1$. For $n = 3, \frac{1}{8} = 0.125 > 0.1$; for $n = 4, \frac{1}{16} = 0.0625 < 0.1$. Hence the least value is $n = 4$.

119. (a)

Solution: Here $p = \frac{1}{3}, q = \frac{2}{3}$. The ratio of consecutive terms is $\frac{P(X=2)}{P(X=1)} = \frac{n-k+1}{k} \cdot \frac{p}{q}$ with $k = 2$: $\frac{6-2+1}{2} \cdot \frac{1/3}{2/3} = \frac{5}{2} \cdot \frac{1}{2} = \frac{5}{4}$.

120. (d)

Solution: Here $n = 4, p = q = \frac{1}{2}$. $P(X = 0) = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$ and $P(X = 4) = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$. Sum $= \frac{2}{16} = \frac{1}{8}$.